

Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

12 V

6 cells

18 Ah

20 h to 10.5 V at 25°C

AGM

Valve-regulated lead-acid

Electrical performance

Rated capacity	18 Ah at 20 h to 10.5 V, 25°C
10-hour capacity	16.8 Ah to 10.5 V at 25°C
1-hour capacity	11 Ah to 9.60 V at 25°C
Internal resistance	15 mohm at 25°C
Maximum discharge	270A for 5 seconds at 25°C
Self-discharge	<=3% per month at 25°C
Design life / standby	5 years (JIS, 25°C); 3-5 years (Eurobat, 20°C)

Charging and operating conditions

Charge method	Constant voltage
Float / cycle voltage	13.6V-13.8V / 14.5V-15.0V at 25°C
Maximum charge current	5.4A
Temperature compensation	Float: -18 mV/°C per battery; cycle: -30 mV/°C per battery
Operating temperatures	Discharge: -15 to +50°C; charge: -10 to +50°C
Storage temperature	-20 to +50°C
Nominal operating temperature	25 ±3°C

Mechanical specification

Dimensions (L × W × H)	181±2 × 77±1 × 167±2 mm; overall height 167±2 mm
Approximate weight	5.20 kg ±4%
Terminal / torque	T3/T12; 3.0±0.6N.m
Container	ABS, UL94-HB; V-0 optional

Construction & applications

Valve-regulated lead-acid battery using AGM (Absorbent Glass Mat) technology, high-performance plates and electrolyte. Designed for standby and general-purpose applications.

Battery construction

Positive plate	Lead dioxide
Negative plate	Lead
Container / lid	ABS / ABS
Safety valve	Rubber
Terminals	Copper
Separator	Glass fibre (AGM)
Electrolyte	Sulfuric acid

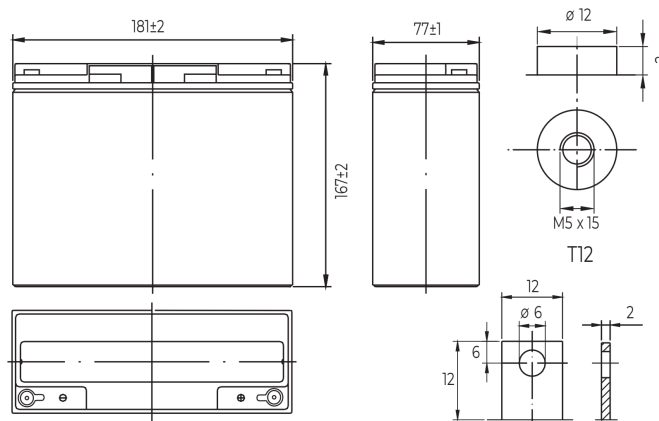
Features

- Valve-regulated lead-acid construction.
- AGM separator technology.
- High-performance plates and electrolyte.
- ABS container: UL94-HB; V-0 optional.
- Self-discharge: up to 3% per month at 25°C.

Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

Discharge performance

Fully charged battery / 25°C

Constant-current discharge / A

V/cell	5m	10m	15m	30m	1h	2h	3h	4h	5h	10h	20h
1.60	68.80	43.60	34.00	19.20	11.80	6.45	4.46	3.69	3.14	1.70	0.91
1.65	66.70	42.30	33.20	18.80	11.60	6.41	4.44	3.67	3.12	1.70	0.91
1.70	63.90	40.50	32.00	18.20	11.30	6.35	4.40	3.64	3.10	1.69	0.91
1.75	61.20	38.80	30.90	17.80	11.10	6.26	4.37	3.62	3.08	1.68	0.90
1.80	57.80	36.60	29.30	17.10	10.80	6.10	4.24	3.51	2.99	1.65	0.88

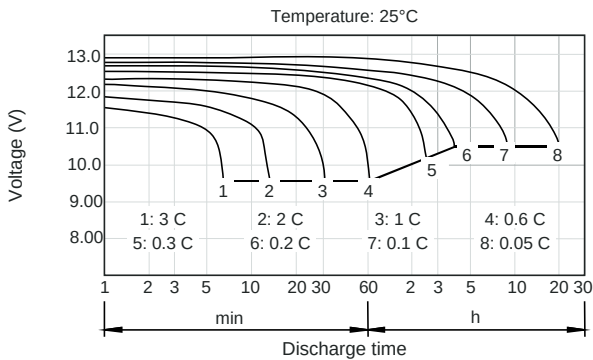
Constant-power discharge / W per cell

V/cell	5m	10m	15m	30m	1h	2h	3h	4h	5h	10h	20h
1.60	127.83	81.83	64.67	36.67	22.83	12.58	8.83	7.33	6.25	3.42	1.83
1.65	124.00	79.50	63.17	35.83	22.33	12.50	8.78	7.28	6.22	3.40	1.82
1.70	119.00	76.17	60.83	34.83	21.83	12.38	8.72	7.23	6.17	3.38	1.82
1.75	113.83	72.83	58.67	34.00	21.33	12.20	8.67	7.18	6.13	3.37	1.80
1.80	107.50	68.83	55.67	32.67	20.83	11.90	8.40	6.97	5.93	3.28	1.77

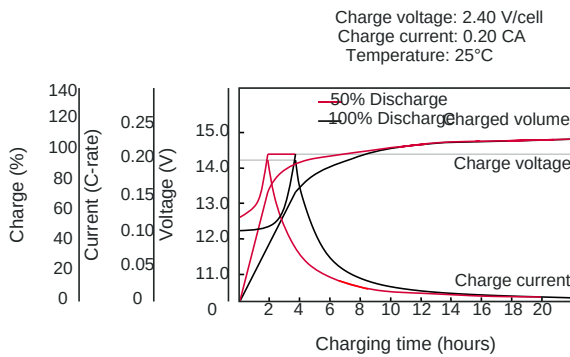
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

Characteristic curves

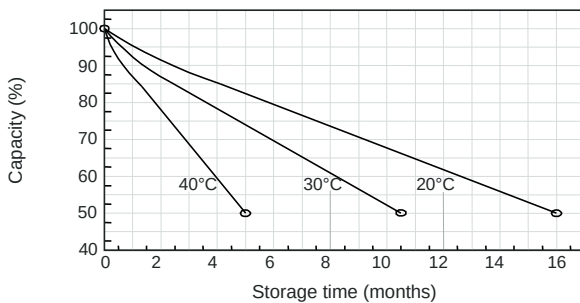
Discharge characteristics



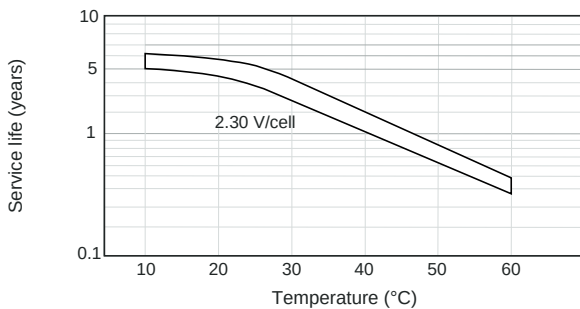
Charging characteristics



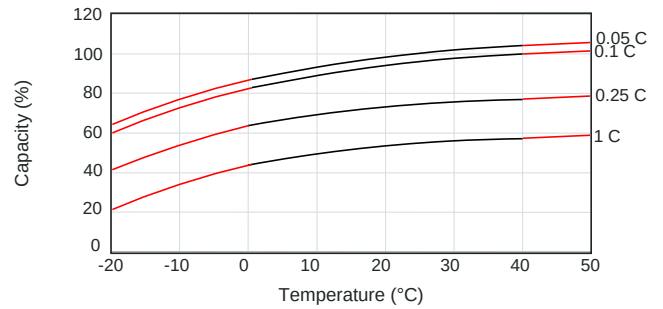
Self-discharge



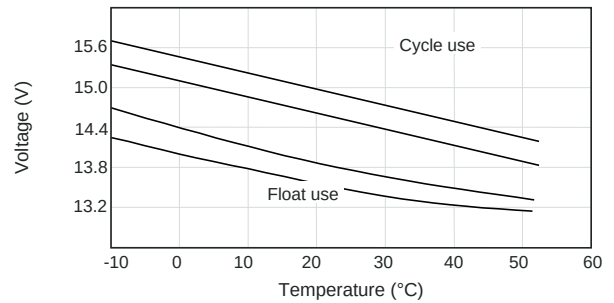
Float life vs. temperature



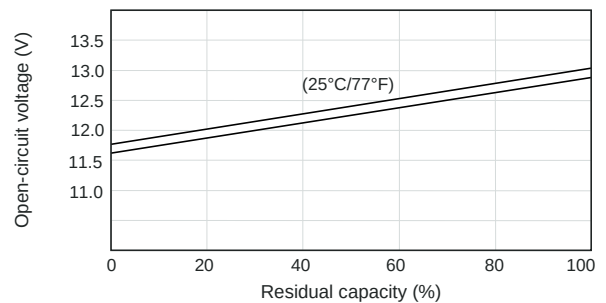
Capacity vs. temperature



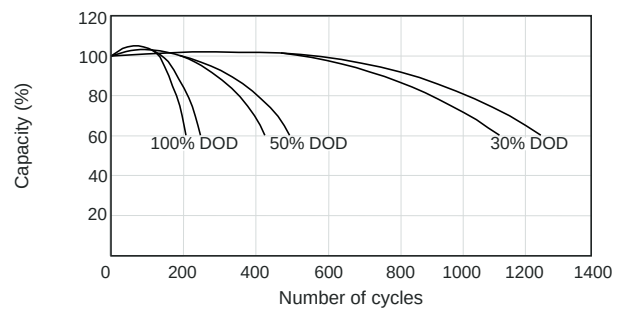
Charge voltage vs. temperature



Open-circuit voltage vs. capacity



Cycle life vs. depth of discharge



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.